CPS School Level Enrollment Projections Methodology

Overview

The enrollment projection system used by Chicago Public Schools to generate 10-year forecasts at the school level. The process begins by integrating GIS geographic data with over a decade of historical enrollment and demographic records to establish a baseline for every school and community area. To project future student populations, the system utilizes three primary methodologies: Cohort Survival Ratios for traditional student persistence, Birth-to-Enrollment rates for entry-level grades, and Average models for specialized school types. A central feature of the system is its automated model optimization, which tests various Simple and Exponential Moving Averages against historical data. By calculating the Root Mean Squared Error, the software identifies and selects the most accurate mathematical fit for each specific grade and location. The final output provides a comprehensive longitudinal analysis, incorporating outlier detection and data exports to assist in long-term strategic district planning.

Data Sources

Database Connectivity: The script connects to GIS databases to pull live lists of schools, community areas, and ARA regions.
Geographic Mapping: Every school is mapped to its specific community area and region. Additionally, community areas and regions are treated as "schools" within the code to allow for aggregate-level projections.
Historical Compilation: The system compiles approximately 11–12 years of historical data from multiple sources, including school-level "Tarantula" files and demographic "20-day" enrollment files. Tarantula files include additional details for Attendance-Area (Zoned) schools. For these schools, population is broken down by students zoned to the school (live within the boundary) and students attending, but living outside of the boundary. Additionally, Total Residing population for students living within the attendance boundary is also included.

Forecasting Methodologies

The system uses three distinct mathematical approaches depending on the grade level and school stability:

1. Cohort Survival Ratio (CSR): Used primarily for "stable" or "Traditional" schools. It calculates the ratio of students in one grade this year compared to those in the preceding grade the previous year to measure student persistence.

Formula: CSR = Current Year Enrollment in Grade X/ Prior Year Enrollment in Grade X-1

CSR is a ratio. Because it is a percentage-based metric, it naturally scales across school sizes. A CSR of 0.95 indicates a 5% loss in students regardless of whether the school has 100 or 1,000 students, making it a more normalized indicator of persistence than absolute numbers.

2. Use Averages (AVG) for Volatile or Small Contexts

While CSR is effective for traditional schools, there is a switch to an **Average (AVG) model** for schools with "unstable" enrollment or non-standard student assignments, such as **Options, Specialty, or Early Childhood schools**

Handling small schools: In very small schools, a single student leaving can cause a massive swing in a CSR ratio (the "small denominator" problem). For these contexts, or for entry grades (PE, PK, K) where no prior grade exists to calculate a ratio, using a moving average of total enrollment can be more reliable.

3. Birth-Based Forecasting: For entry grades (PE, PK, K), the model uses birth data from the Illinois Department of Public Health (ILDPH). It runs a **linear regression** to estimate future births and then calculates a "Birth-to-Enrollment" rate to project future district-wide early childhood populations.

Automated Model Selection via RMSE

Rather than applying a "one-size-fits-all" average, the system tests multiple variations and selects the best one for each specific school and grade. The variations are based on applying the CSR averages, as well as grade-level enrollment averages. The sources achieve this by calculating the **Root Mean Squared Error (RMSE)** for several models:

- **Simple Moving Averages (SMA):** Unweighted averages of the last 1 to 5 years
- **Exponential Moving Averages (EMA):** Weighted averages using **alpha values** from 0.05 to 0.5, which prioritize more recent data

The script calculates the **Root Mean Squared Error (RMSE)** for every variation by comparing what the model *would have* predicted in the past against what *actually* happened. The variation with the **lowest RMSE** is automatically selected for the future projection (e.g., a 3-year SMA vs. a 0.2 alpha EMA).

$$RMSE=SQRT(\sum(\text{Actual Enrollment}-\text{Predicted Enrollment})^2/n)$$

Handling Incomplete Data

The system includes logic to handle schools or categories with limited historical records. It determines how many years of data are available for a specific school-grade combination.

- **Full Data:** If 10 or more years of data are present, the system compares all SMA and EMA variations.
- **Incomplete Data:** If fewer years are available (e.g., "incompletecount3"), the system restricts its selection to simpler models, such as a 2-year SMA, because there is insufficient history to calculate a reliable multi-year EMA.
- **Single-Year Data:** If only one year of data exists, the system defaults to **the previous year's data.**

Automated Guardrails

Once the model with the lowest RMSE is chosen, the system applies further "guardrails" during the projection phase. If the selected model results in an outlier—such as a CSR below 0.8 or above 1.2—the projection is flagged for manual review to ensure the automated RMSE selection hasn't picked a mathematically accurate but realistically improbable trend.

Executing Multi-Year Projections

Once the "best-fit" model (CSR or AVG) is selected, the script generates a **10-year forecast**.
- **CSR Projections:** The Year 1 projection is calculated as (Current Enrollment×Chosen CSR). Subsequent years use the previous year's projection as the new base, with the chosen CSR.
- **AVG Projections:** The chosen historical average is applied consistently across all forecast years.

Subgroups

The system is designed to handle demographic subgroups. The default setting is set to 'All Students' but can be modified to include specific subgroups (ie. Asian, Latino, Special Education, etc.). By default, the residing population for the 77 community areas and 16 ARA Regions are included. These are treated as individual schools.

Once a subgroup is selected, the system continues to apply its standard logic—such as testing for the lowest RMSE and choosing between CSR or Enrollment Averaging models—to that specific population.

Wishlist - Issues:

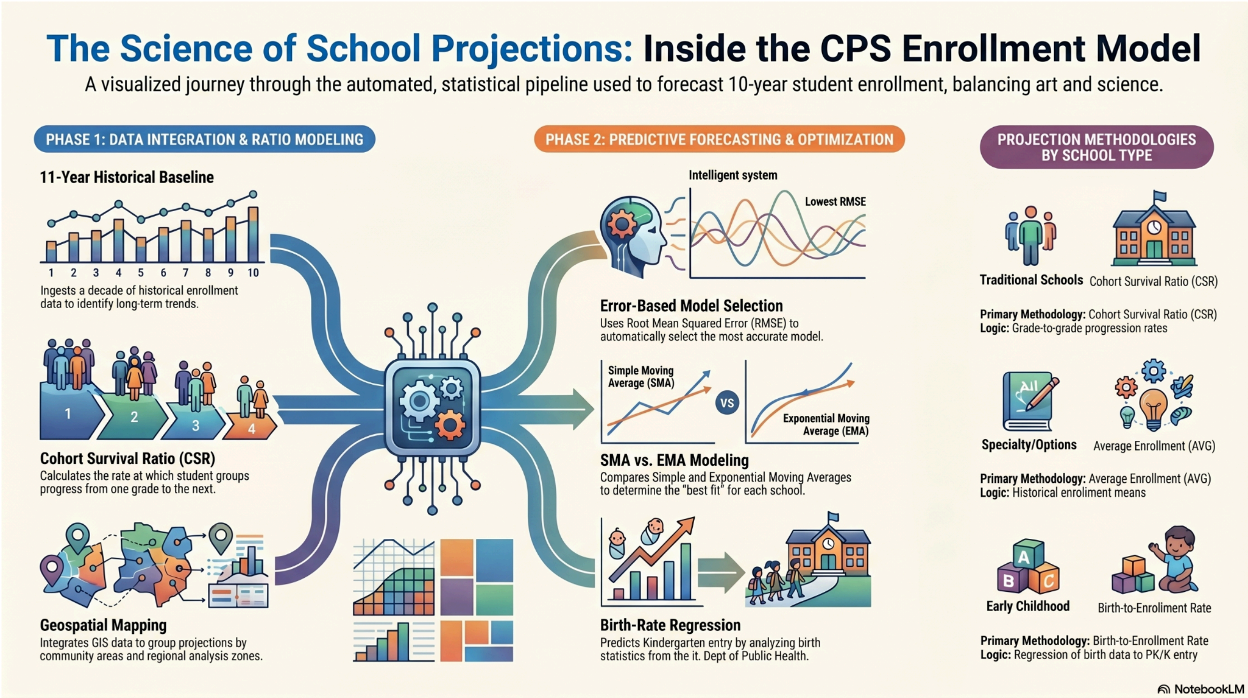
- The school district has a large number of small schools – these seem to have less reliable results since enrollment for a particular grade or subset of students may fluctuate more from one year to the next. It's possible schools below a certain enrollment threshold need to be treated differently.

- We have not validated the models used based on existing data, to determine adjustments that still need to be made. We may be able to group schools based on specific characteristics to adjust the model.
- Schools may increase or decrease in enrollment based on population changes. Many areas of the city are declining in enrollment due to population loss and lower birth rates. Demographic changes are taken into account for CSRs, however, the model creates averages for entry grades which won't be accurate for schools with increasing or decreasing entry-level grades. This is one of the bigger issues because it impacts the majority of the schools.

Other data available:

- We have data on where students live within the district. Data shared is only for students zoned to a school, but we could provide information on students living in the area, for example # of 8th graders that will be zoned to the high school the following year (assuming they don't move)
- Data on students moving (out of the district, or to another part of the city) or students transferring schools.
- Data on applications for 9th grade - The majority of students apply to 9th grade, data includes applications, offers, and acceptances.

As of May 2026


The Science of School Projections: Inside the CPS Enrollment Model
A visualized journey through the automated, statistical pipeline used to forecast 10-year student enrollment, balancing art and science.
PHASE 1: DATA INTEGRATION & RATIO MODELING
11-Year Historical Baseline
1 2 3 4 5 6 7 8 9 10
Ingests a decade of historical enrollment data to identify long-term trends.
1 2 3 4
Cohort Survival Ratio (CSR)
Calculates the rate at which student groups progress from one grade to the next.
Geospatial Mapping
Integrates GIS data to group projections by community areas and regional analysis zones.
PHASE 2: PREDICTIVE FORECASTING & OPTIMIZATION
Intelligent system
Lowest RMSE
Error-Based Model Selection
Uses Root Mean Squared Error (RMSE) to automatically select the most accurate model.
Simple Moving Average (SMA)
VS
Exponential Moving Average (EMA)
SMA vs. EMA Modeling
Compares Simple and Exponential Moving Averages to determine the "best fit" for each school.
Birth-Rate Regression
Predicts Kindergarten entry by analyzing birth statistics from the it. Dept of Public Health.
PROJECTION METHODOLOGIES BY SCHOOL TYPE
Traditional Schools
Cohort Survival Ratio (CSR)
Primary Methodology: Cohort Survival Ratio (CSR)
Logic: Grade-to-grade progression rates
Specialty/Options
Average Enrollment (AVG)
Primary Methodology: Average Enrollment (AVG)
Logic: Historical enrollment means
A B C
Early Childhood
Birth-to-Enrollment Rate
Primary Methodology: Birth-to-Enrollment Rate
Logic: Regression of birth data to PK/K entry
NotebookLM